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Assignment 8

Numerical Integration Using the Trapezoidal Rule, Simpson's 1/3, Simpson's 3/8th Rule

1. Numerical Integration of $\int_0^2 x^2 dx$

Problem: Evaluate the integral $I = \int_0^2 x^2 dx$ using the Trapezoidal Rule, Simpson's 1/3 Rule, and Simpson's 3/8th Rule.

Exact Value: $I = [x^3/3]$ (evaluated from 0 to 2) $= 8/3 = 2.6666667$

Trapezoidal Rule: $I_t = (h/2) [f(x_0) + 2\sum f(x_i) + f(x_n)]$ where $h = (b - a)/n = 2/n$ and $f(x) = x^2$.

Simpson's 1/3 Rule: $I = (h/3) [f(x_0) + 4\sum f(x_i) + 2\sum f(x_i) + f(x_n)]$, where the coefficients 4 and 2 are applied alternately to the interior points.

Simpson's 3/8th Rule: $I = (3h/8) [f(x_0) + 3\sum f(x_i) + 2\sum f(x_i) + f(x_n)]$, where the coefficients 3 and 2 are applied according to the Simpson's 3/8 pattern.

Absolute Error: $\text{Error} = |I - I_{\text{approx}}|$

n	Trapezoidal Approximation	Trapezoidal Absolute Error	Simpson's 1/3 Approximation	Simpson's 1/3 Absolute Error	Simpson's 3/8 Approximation	Simpson's 3/8 Absolute Error
1	4.000000	1.333333	—	—	—	—
2	3.000000	0.333333	2.666667	0.000000	—	—
3	—	—	—	—	2.666667	0.000000
4	2.750000	0.083333	2.666667	0.000000	—	—
6	—	—	—	—	2.666667	0.000000
8	2.687500	0.020833	2.666667	0.000000	—	—
12	—	—	—	—	2.666667	0.000000
16	2.671875	0.005208	2.666667	0.000000	—	—

Example for $n = 2$:

$$h = (2 - 0)/2 = 1$$

$$I_t = (1/2)[f(0) + 2f(1) + f(2)] = (1/2)[0 + 2(1) + 4] = 3$$

$$\text{Error} = |8/3 - 3| = 1/3 = 0.333333$$

2. A Function Without an Elementary Antiderivative

Consider $I = \int_0^1 e^{-x^2} dx$. Use the composite trapezoidal rule, Simpson's 1/3 Rule, and Simpson's 3/8th Rule.

Reference value for comparison: $I \approx 0.7468241328$.

n	Trapezoidal Approximation	Trapezoidal Absolute Error	Simpson's 1/3 Approximation	Simpson's 1/3 Absolute Error	Simpson's 3/8 Approximation	Simpson's 3/8 Absolute Error
3	—	—	—	—	0.746824691	0.000000558
4	0.7429840978	0.0038400350	0.746833604	0.000009471	—	—
6	—	—	—	—	0.746825062	0.000000929
8	0.7458656148	0.0009585180	0.746824175	0.000000042	—	—
12	—	—	—	—	0.746824188	0.000000055
16	0.7465845968	0.0002395360	0.746824135	0.000000002	—	—
24	—	—	—	—	0.746824134	0.000000001
32	0.7467642547	0.0000598782	0.746824133	0.000000000	—	—

Note: Simpson's 1/3 Rule requires an even number of subintervals, while Simpson's 3/8th Rule requires the number of subintervals to be a multiple of 3.

3. Data Instead of a Function

Given the experimental data:

x	0	0.5	1.0	1.5	2.0
f(x)	1.00	1.65	2.70	4.48	7.39

Here $h = 0.5$.

Applying the composite trapezoidal rule: $It = (0.5/2)[1.00 + 2(1.65 + 2.70 + 4.48) + 7.39]$

$It = 6.5125$

Applying Simpson's 1/3 Rule to all four subintervals: $I = (0.5/3)[1.00 + 4(1.65 + 4.48) + 2(2.70)] = 6.5016667$

Absolute error: Not available from the given assignment data because no exact/reference value is provided.

Simpson's 3/8th Rule cannot be applied directly to all four subintervals because the number of subintervals is not a multiple of 3. Using Simpson's 3/8th Rule for the first three subintervals and the trapezoidal rule for the remaining interval:

$$I_{3/8} = (3 \times 0.5/8)[1.00 + 3(1.65) + 3(2.70) + 4.48] = 3.474375$$

$$I_{\text{trap}} = (0.5/2)[4.48 + 7.39] = 2.9675$$

$$\text{Total approximation} = 3.474375 + 2.9675 = 6.441875$$

Absolute error: Not available from the given assignment data because no exact/reference value is provided.